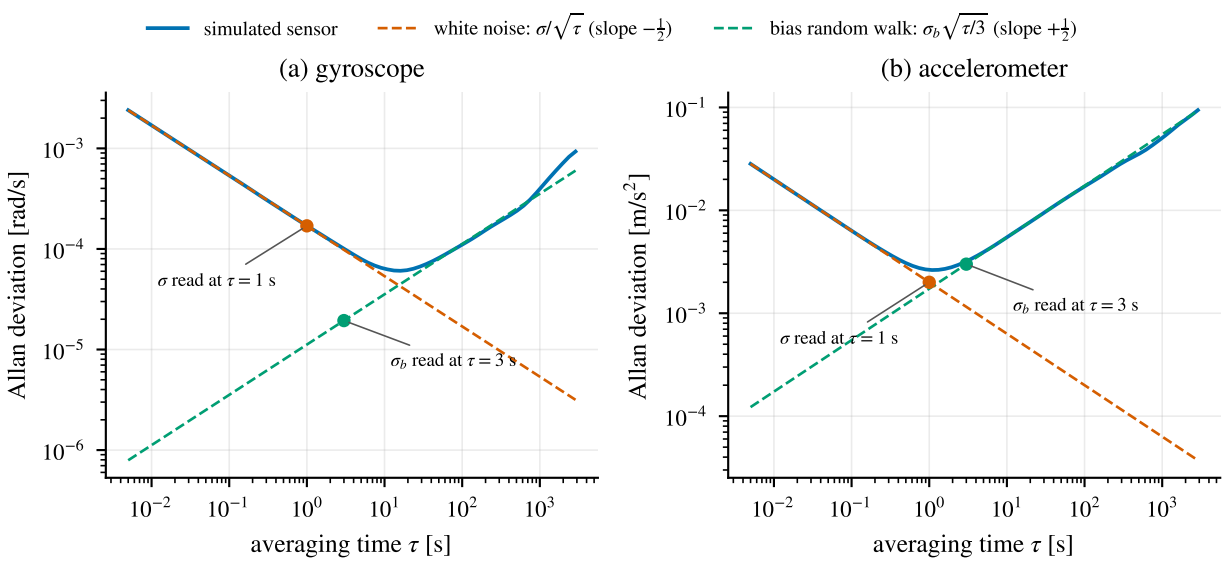
simulated sensor
white noise: $\sigma/\sqrt{\tau}$ (slope $-\frac{1}{2}$)
bias random walk: $\sigma_b\sqrt{\tau/3}$ (slope $+\frac{1}{2}$)
(a) gyroscope
Allan deviation [rad/s]
10^{-3}
10^{-4}
10^{-5}
10^{-6}
σ read at $\tau = 1$ s
σ_b read at $\tau = 3$ s
10^{-2} 10^{-1} 10^{0} 10^{1} 10^{2} 10^{3}
averaging time τ [s]
(b) accelerometer
Allan deviation [m/s^2]
10^{-1}
10^{-2}
10^{-3}
10^{-4}
σ_b read at $\tau = 3$ s
σ read at $\tau = 1$ s
10^{-2} 10^{-1} 10^{0} 10^{1} 10^{2} 10^{3}
averaging time τ [s]